TRANSFORMS AND PDE. 24MAT209

Syllabus

UNIT 1:

Fourier Series: Even & Odd f^n, Half Range expansions; 11.1-11.4

UNIT 2:

& Sturm-Liouville: Generalised fourier series, fourier integral and fourier transforms; 11.5-11.10.

UNIT 3:

Basic Concept of PDE: Solution by seperating variables, D'Alembert's solution of wave eqn; 12.1-12.4

UNIT 4:

Heat eq^n; Solution by fourier series, Heat eqn for long bar, Solution by fourier integrals and transforms. 2-D wave eq^n; 12.5-12.8

UNIT 5:

Laplace in polar coordinates, Fourier-Bessel's Series, Laplace eq^n by cylindrical and spherical coordinates, Solutions of PDE by Laplace transforms. 12.8-12.12.

Text book

① ERWIN KREYSEG; Advanced engineering Mathematics: 10

② DENNIS ZIMM; A first course in diff eq^n: Cengage

Reference

① WE Boyce, Elementary differential eq^n & Boundry value probles: 7 edi

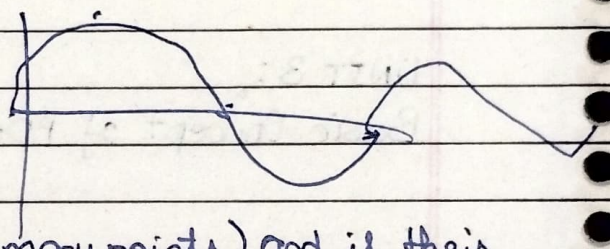
② R.C Dprima

Fourier Series

- Mathematical technique used to express any periodic functions as an infinite sum of sin's & cosine f^n's.

Application: - It helps to solve diff eq^n especially boundry valued problems, analyse vibrations of strings, study heat eq^n, remove noise from signals.

Periodic



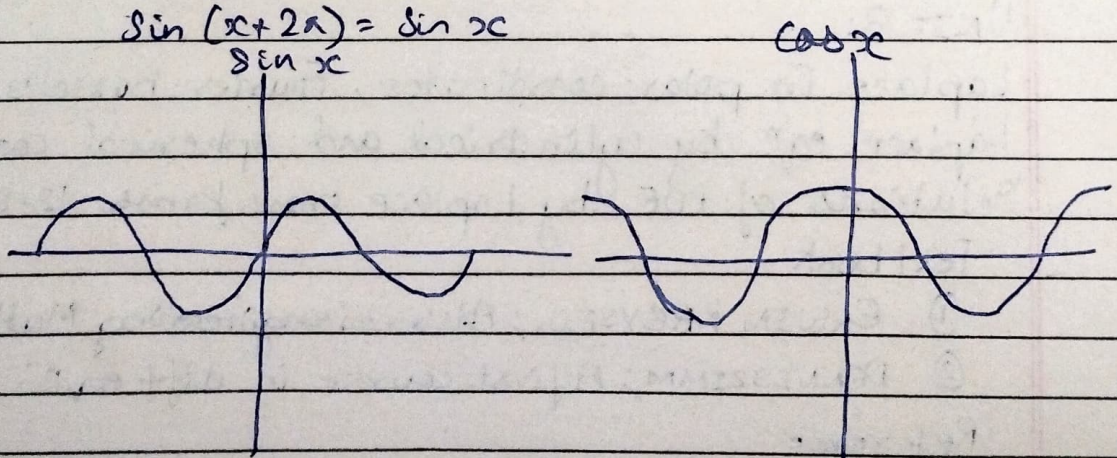
A function $f(x)$ is said to be a periodic function if $f(x)$ is defined for all values of $x \in R$ (except at countably many points) and if there is a P (real) such that

$f(x+P) = f(x), \forall x \in R$, the +ve no P is called period of the f^n $f(x)$.

eg: $\sin x$, $\cos x$ are periodic with 2π

$\cos(x+2\pi) = \cos x$

$\sin(x+2\pi) = \sin x$



Note: $\cos x$, $\sin x$ has also period $2\pi, 4\pi, 6\pi \ldots$ in this the smallest period 2π is called fundamental Period.

* Polynomial function, exponential, log all are non periodic.

REMARK 1

- If $f(x)$ have a period P then $2P, 3P, 4P, \ldots, nP$ are also period of $f(x)$

given $f(x+P) = f(x)$, $\forall\, x \in R$

$f(x+2P) = f(x+P+P) = f(x+P) = f(x)$

$\vdots$

$f(x+nP) = f(x)$ where $n = 1, 2, 3 \ldots$

REMARK 2

- If $f(x)$, $g(x)$ have period P, then their linear combination have period P.

given: $f(x+P) = f(x)$ & $g(x+P) = g(x)$

let $h(x) = af(x) + bg(x)$

$h(x+P) = af(x+P) + b(g(x+P)$

$= af(x) + b(g(x)$

$h(x+P) = h(x)$

REMARK 3

- If $f(x)$ is a est $f(x) = K$ is periodic with any period but it does not have a fundamental period.

RESULT

If $f(x)$ has period P, then $f(ax)$ has period $\frac{P}{a}$ and $f(x/b)$ has period BP.

$f(a(x + \frac{P}{a}) = f(ax+P) = f(ax)$ $\therefore$ period $= \frac{P}{a}$

$f\left(\frac{2x+bP}{b}\right) = f\left(\frac{x}{b}+P\right) = f\left(\frac{x}{P}\right)$

Q. Find period of

a) $\sin 3x$

$= \frac{2\pi}{3}$

b) $\sin \frac{2x}{L}$

$\frac{2\pi}{2/L} = L\pi$

c) $\cos \frac{4x}{8L}$

$= \frac{2\pi}{4/8L} = 4\pi L$

d) $\cos \frac{x}{100}$

$= \frac{2\pi}{\frac{1}{100}} = 200\pi$

e) $\cos \frac{3x}{100}$

$= \frac{2\pi}{\frac{3}{100}} = \frac{200\pi}{3}$

Q. Find a periodic $f(x)$ whose period is 5.
either $\sin \frac{2x\pi}{5}$ or $\cos \frac{2x\pi}{5}$

Trigonometric Series

$$S = \{a_0, a_1, a_2 \ldots a_n \ldots\}$$

is called an infinite sequence, then sum of all terms of S i.e

$$a_0 + a_1 + \cdots + a_n + \cdots = \sum_{n=0}^{\infty} a_n$$

is called an infinite series.

A trigonometric series is a series of the form

$$a_0 \cos 0x + b_0 \sin 0x + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \cdots + a_n \sin nx + a_n \cos nx + \cdots$$

in sigma notation

$$a_0 \cos 0x = 1 \times a_0$$

$$b_0 \sin 0x = 0$$

$$\Rightarrow \boxed{a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx}$$

→ Consider the following

	Period
$f(x) = 1$	any +ve real
$f(x) = \cos x$	$2\pi, 4\pi, 6\pi \ldots$
$f(x) = \sin x$	$2\pi, 4\pi, 6\pi \ldots$
$f(x) = \cos 2x$	$\pi, 2\pi, 3\pi \ldots$
$f(x) = \sin 2x$	$\pi, 2\pi, 3\pi \ldots$
$f(x) = \cos 3x$	$\frac{2\pi}{3}, \frac{4\pi}{3}, \frac{6\pi}{3} = 2\pi, \ldots$
$f(x) = \sin 3x$	$\frac{2\pi}{3}, \frac{4\pi}{3}, \frac{6\pi}{3} = 2\pi \ldots$

→ Since all these had common period 2π their linear combination will also have period 2π.

→ If $f(x)$ is known, this $f(x)$ can be expanded as a sum of trigonometric series called the Fourier series of $f(x)$.

$$f(x) = a_0 + \left(\sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx\right) \quad \text{— (1)}$$

where a_0, a_n, b_n are called the Fourier coeff and evaluated by euler formulas.

→ If $f(x)$ is periodic with period 2π the sum of the trignometric series i.e (RHS of eqn (1)) converges to $f(x)$.

→ The converges depends on the value of a_0, a_n, b_n.

a) $\int_{-\pi}^{\pi} \cos nx \cos mx \, dx$

~~$\cos(n+m)x = \cos nx + \cos mx - \sin nx \sin mx$~~

$\cos(n+m) = \cos n \cos m - \sin n \sin m$

$\cos(n-m) = \cos n \cos m + \sin n \sin m$

$\cos(n+m) + \cos(n-m) = 2 \cos n \cos m$

$\cos n \cos m = \frac{1}{2}\left[\cos(n+m) + \cos(n-m)\right]$

$$\int_{-\pi}^{\pi} \cos nx \cos mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} \cos(n+m)x + \cos(n-m)x$$

$$= \frac{1}{2}\left[\frac{\sin(n+m)}{n+m} + \frac{\sin(n-m)}{n-m}\right]_{-\pi}^{\pi}$$

$= 0, \; n \neq m$

when $n=m$

$$= \int_{-\pi}^{\pi} \cos^2 mx \, dx$$

$$= \int_{-\pi}^{\pi} \left(\frac{1+2\cos 2mx}{2}\right) dx$$

$$= \frac{1}{2}\left[x + \frac{\sin 2mx}{2m}\right]_{-\pi}^{\pi}$$

$$\frac{1}{2}\left[2\pi\right]$$

$$= \pi$$

b) $$\int_{-\pi}^{\pi} \sin nx \sin mx \, dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi} \cos(n-m)x - \cos(n+m)x \, dx$$

$$= \frac{1}{2}\left[\frac{\sin(n-m)x}{n-m} - \frac{\sin(n+m)x}{n+m}\right]_{-\pi}^{\pi}$$

$= 0, \; n \neq m$

If $n=m$

$$\int_{-\pi}^{\pi} \sin^2 nx \, dx = \int_{-\pi}^{\pi} \frac{1-\cos 2nx}{2} \, dx$$

$$= \frac{1}{2}\left[x - \frac{\sin 2mx}{2m}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{2} \times 2\pi = \pi$$

c) $\int_{-\pi}^{\pi} \sin nx \cos mx \, dx$

$\int_{-\pi}^{\pi} \sin(n+m)x + \sin(n-m)x \, dx$

$= -\frac{1}{2}\left[\frac{\cos(n+m)x}{n+m} + \frac{\cos(n-m)x}{(n-m)}\right]_{-\pi}^{\pi}$

$= 0$ (when $n \neq m$)

when $n=m$

$\int_{-\pi}^{\pi} \sin mx + \cos nx = \int_{-\pi}^{\pi} \sin 2mx \, dx$

$= \frac{1}{2}\left[\frac{\cos 2mx}{2m}\right]_{-\pi}^{\pi}$

$= 0$

Euler Formula

Let $f(x)$ is a periodic f^n with period 2π, there the fourier series of $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$ —(1)

(a) integrate eq^n (1) from $-\pi$ to π

$\int_{-\pi}^{\pi} f(x)\,dx = \int_{-\pi}^{\pi} a_0 \, dx + \int_{-\pi}^{\pi} \left(\sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx\right) dx$

$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$

To find a_n multiply eqn ① by $\cos mx$ and integrate from $-\pi$ to π

$$\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \int_{-\pi}^{\pi} a_0 \cos mx\,dx + \sum_{n=1}^{\infty}\left(\int_{-\pi}^{\pi} a_n \cos nx\,dx \cos mx\,dx + \int_{-\pi}^{\pi} b_n \cos nx\,dx \sin mx\,dx\right)$$

$$= a_0 \times 0 + a_m \pi + b_n \times 0$$

$$a_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos mx\,dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

multiply ① $\sin mx$ and integrate from $-\pi$ to π

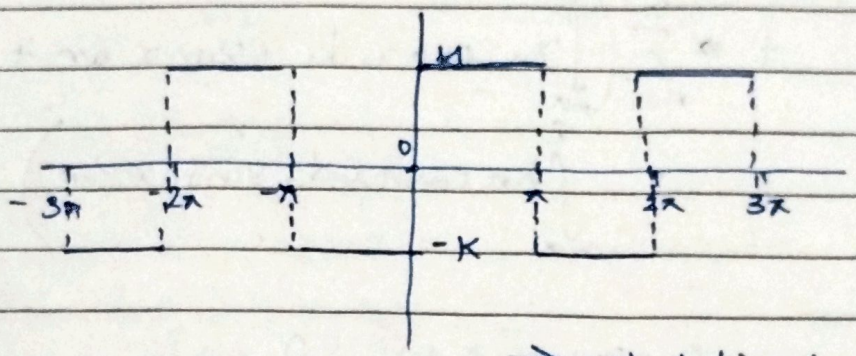
$$\int_{-\pi}^{\pi} f(x)\sin mx\,dx = \int_{-\pi}^{\pi} a_0 \sin mx\,dx + \sum_{m=1}^{\infty}\left(\int_{-\pi}^{\pi} a_n \cos nx \sin mx\,dx + \int_{-\pi}^{\pi} b_n \sin nx \sin mx\,dx\right)$$

$$= 0 + a_n \times 0 + b_n \times \pi$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cdot \sin(nx)\,dx$$

Draw

① $f(x) = \begin{cases} -K, & -\pi < x < 0 \\ K, & 0 < x < \pi \end{cases}$, $f(x+2\pi) = f(x)$



→ not defined at $-\pi, 0, \pi$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{0} -K\, dx + \int_{0}^{\pi} K\, dx$$

$$= \frac{K}{\pi} \left[(-x)_{-\pi}^{0} + (x)_{0}^{\pi} \right]$$

$$= \frac{K}{\pi} (\pi + \pi) = \underline{0}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{0} -K \cos nx + \int_{0}^{\pi} K \cos nx\, dx$$

$$a_n = \frac{1}{\pi} \left[\left(\frac{-\sin nx}{n} \right)_{-\pi}^{0} + \left(\frac{\sin nx}{n} \right)_{0}^{\pi} \right]$$

$$\underline{a_n = 0}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} k\sin nx\,dx + \int_{0}^{\pi} k\sin nx\,dx\right]$$

$$= \frac{k}{\pi}\left[\left(\frac{\cos nx}{n}\right)_{-\pi}^{0} - \left(\frac{\cos nx}{n}\right)_{0}^{\pi}\right]$$

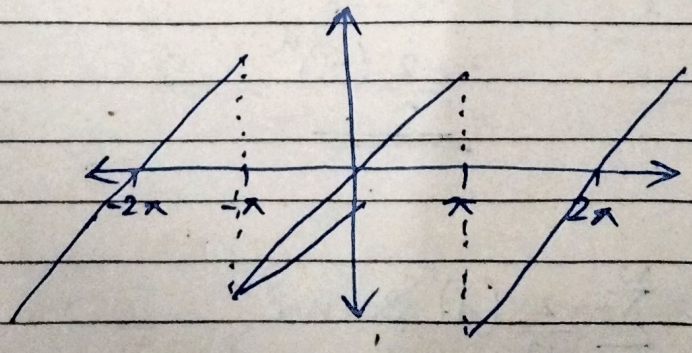
$$= \frac{k}{n\pi}\left[1 - \cos n\pi - \cos n\pi + 1\right]$$

$$= \frac{2k}{n\pi}\left[1 - \cos n\pi\right]$$

$$b_n = \frac{2k}{n\pi}\left[1 - (-1)^n\right]$$

$$b_n = \begin{cases} 0 & \text{if } n \text{ is even} \\ \frac{4k}{n\pi} & \text{if } n \text{ is odd} \end{cases}$$

H.W $f(x) = x$, $-\pi < x < \pi$, $f(x+2\pi) = f(x)$



$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$f(x) = x$

$f(-x) = -x$

(odd f^n)

$a_0 = 0$

$\int_{-a}^{a} f(x)\,dx = 0$ ↳ odd f^n

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

$-x\cos n(x)$

$= -x\cos nx$

$-f(x) =$ odd

$a_n = 0$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$x \to$ odd $-ve$

$\sin x \to$ odd $-ve$

$x\sin x \to$ even $+ve$

$\int_{-\pi}^{\pi} x\sin x\,dx = 2\int_{0}^{\pi} x\sin x\,dx$ (even)

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} x\cdot\sin nx\,dx$$

$$= \frac{1}{\pi}\left[2\int_{0}^{\pi} x\sin nx\,dx\right]$$

$$= \frac{2}{\pi}\left[\frac{-x\cos nx}{n} + \frac{x\sin nx}{n^2}\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{-\pi}{n}\cos(n\pi)\right] = \frac{-2}{n}\cos n\pi$$

$$= \frac{-2}{n}(-1)^n$$

F.S =

$$f(x) = \sum_{n=1}^{\infty} \frac{-2}{n}(-1)^n \sin nx$$

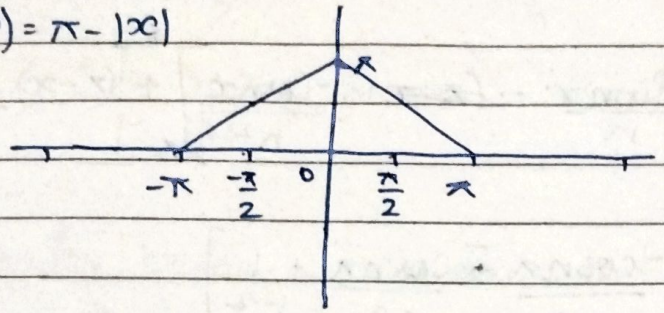
Q. Draw the periodic graph

1) $f(x) = \pi - |x|$, $-\pi < x < \pi$, $f(x+2\pi) = f(x)$

2) $f(x) = e^{-|x|}$, $-\pi < x < \pi$

then find its fourier series.

1) $f(x) = \pi - |x|$



x	$\pi - \lvert x \rvert$
$-\pi$	0
$-\pi/2$	$\pi/2$
0	π
$\pi/2$	$\pi/2$
π	0

~~2) $f(x) = e^{-|x|}$~~

$$f(x) = \begin{cases} \pi + x & -\pi < x < 0 \\ \pi - x & 0 < x < \pi \end{cases}$$

$$a_0 = \frac{1}{2\pi}\left[\int_{-\pi}^{0} \pi + x\,dx + \int_{0}^{\pi} \pi - x\,dx\right]$$

$$= \frac{1}{2\pi}\left[\left(\pi x + \frac{x^2}{2}\right)_{-\pi}^{0} + \left(\pi x - \frac{x^2}{2}\right)_{0}^{\pi}\right]$$

$$= \frac{1}{2\pi}\left[-\left(-\pi^2 + \frac{\pi^2}{2}\right) + \left(\pi^2 - \frac{\pi^2}{2}\right)\right]$$

$$= \frac{1}{2\pi}\left[\pi^2 - \frac{\pi^2}{2} + \pi^2 - \frac{\pi^2}{2}\right]$$

$$= \frac{1}{2\pi}\left[2\pi^2 - \pi^2\right] = \frac{\pi}{2}$$

FISMIDFIS

$$a_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} f(x)\cos nx\,dx + \int_{0}^{\pi} f(x)\cos nx\,dx\right]$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} (\pi+x)\cos nx\,dx + \int_{0}^{\pi} (\pi-x)\cos nx\,dx\right]$$

$$= \frac{1}{\pi}\left[(\pi+x)\frac{\sin nx}{n} - 1\cdot\frac{-\cos nx}{n^2}\right]_{-\pi}^{0} + \left[(\pi-x)\frac{\sin nx}{n} - (-1)\frac{\cos nx}{n^2}\right]_{0}^{\pi}$$

$$= \frac{1}{\pi}\left[\frac{1}{n^2} - \frac{\cos n\pi}{n^2} + \frac{\cos n\pi}{n^2} + \frac{1}{n^2}\right]$$

$$= \frac{1}{\pi}\left[\frac{2}{n^2} - 2\frac{\cos n\pi}{n^2}\right]$$

$$\frac{2}{n^2\pi}\left[1-\cos n\pi\right]$$

$$= \frac{2}{n^2\pi}\left(1-(-1)^n\right)$$

when n = even $= 0$

when n = odd $\frac{4}{n^2\pi}$

Example bernouli

$$\int x^4 \sin 2x\, dx$$

$$= x^4 x \frac{-\cos 2x}{2} - 4x^3 x \frac{-\sin 2x}{4} + 12x^2 * \frac{\cos 2x}{2^3} - 24x \frac{+\sin 2x}{2^4}$$

$$+ 24x \frac{-\cos 2x}{2^5}$$

→ for bernouli we have to have algebraic first

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b_n of last qv

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx\, dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^{0} (\pi + x) \sin nx\, dx + \int_{0}^{\pi} (\pi - x) \sin nx\, dx \right]$$

$$= \frac{1}{\pi} \left[(\pi+x) \frac{-\cos nx}{n} - 1x \frac{-\sin nx}{n^2} \right]_{-\pi}^{0} + \left[(\pi - x) x \frac{-\cos nx}{n} - (-1) x \frac{-\sin nx}{n^2} \right]_{0}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{-\pi}{n} - \frac{\cancel{\sin 0}}{n^2} + \left(\frac{\pi}{n} \right) \right] = 0$$

Q. Find the fourier series of $f(x) = x^2$ $0 < x < 2\pi$

$f(x+2\pi) = f(x)$

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(x)\,dx$$

$$= \frac{1}{2\pi}\int_0^{2\pi} x^2\,dx$$

$$= \frac{1}{2\pi}\left[\frac{x^3}{3}\right]_0^{2\pi}$$

$$= \frac{1}{2\pi} \times \frac{2\pi \times (2\pi)^2}{3}$$

$$= \frac{4\pi^2}{3}$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x) \cos nx\,dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} x^2 \cos nx\,dx$$

$$= \frac{1}{\pi}\left[x^2 \frac{\sin nx}{n} - 2x \cdot -\frac{\cos nx}{n^2} + 2 \cdot -\frac{\sin nx}{n^3}\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[\frac{x^2 \sin nx}{n} + \frac{2x\cos nx}{n^2} - \frac{2\sin nx}{n^3}\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[\frac{4\pi}{n^2}\right] = \frac{4}{n^2}$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx \Rightarrow \frac{1}{\pi}\int_0^{2\pi} x^2 \sin nx\,dx$$

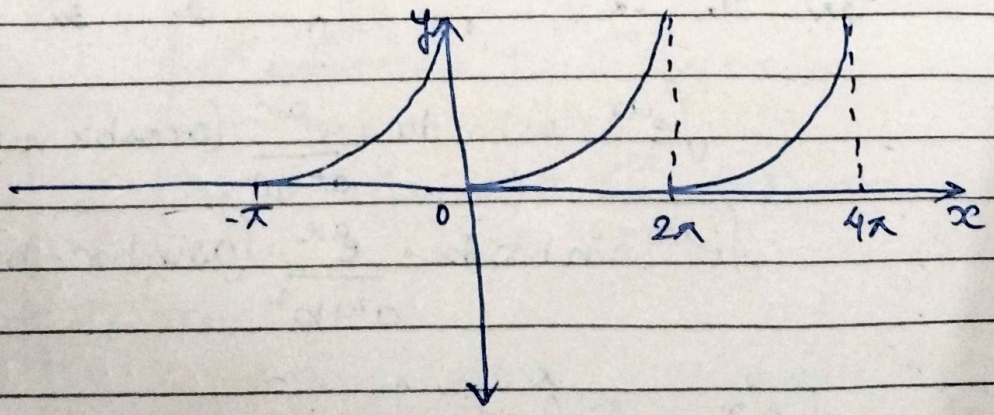
$$= \frac{1}{\pi}\left[\frac{x^2 x - \cos nx}{n} - \frac{2x\,x - \sin nx}{n^2} + \frac{2\cos nx}{n^3}\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[-\frac{x^2\cos nx}{n} + \frac{2x\sin nx}{n^2} + \frac{2\cos nx}{n^3}\right]_0^{2\pi}$$

$$b_n = \frac{1}{\pi}\left(\frac{-4\pi^2}{n} + \frac{2}{n^3} - \frac{2}{n^3}\right)$$

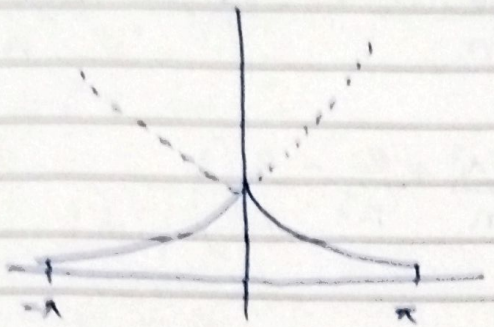
$$= \frac{-4\pi}{n}$$

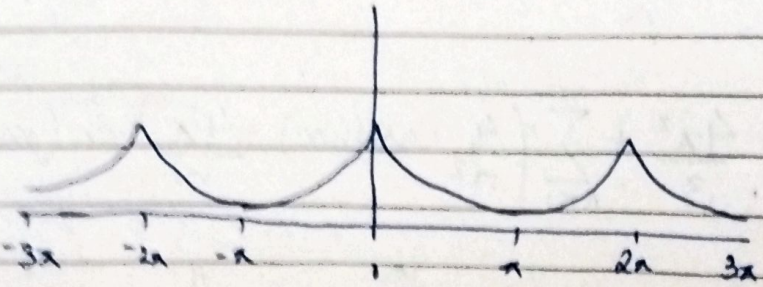
$$x^3 = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty}\left(\frac{4}{n^2}\cos(nx) - \frac{4\pi}{n}\sin(nx)\right) \quad 0<x<2\pi$$



ii) $f(x) = e^{-|x|}$, $-\pi < x < \pi$, $f(x+2\pi) = f(x)$

$$f(x) = \begin{cases} e^{x} & -\pi < x < 0 \\ e^{-x} & 0 < x < \pi \end{cases}$$





$$\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx)$$

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)$$

$$a_0 = \frac{1}{a^2\pi}\left[\int_{-\pi}^{0} e^{x}\,dx + \int_{0}^{\pi} e^{-x}\,dx\right]$$

$$\frac{1}{a^2\pi}\left[e^{x}\right]_{-\pi}^{0} + \left[-e^{-x}\right]_{0}^{\pi}$$

$$\frac{1}{a^2\pi}\left(1 - e^{-\pi} - e^{-\pi} + 1\right)$$

$$\frac{-e^{-\pi}+1}{2\pi} // = \frac{1-e^{-\pi}}{\pi}$$

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$$a_n = \frac{1}{\pi}\int_{-\pi}^{0} e^x \cos nx\,dx + \int_{0}^{\pi} e^{-x}\cos nx\,dx$$

$$= \frac{1}{\pi}\left[\left(\frac{e^x}{1+n^2}(\cos nx + n\sin nx)\right)_{-\pi}^{0} + \left(\frac{e^{-x}}{1+n^2}(-\cos nx + n\sin nx)\right)_{0}^{\pi}\right]$$

$$= \frac{1}{\pi}\left[\left(\frac{1}{1+n^2} - \frac{e^{-\pi}}{1+n^2}\cos n\pi\right) + \left(\frac{e^{-\pi}}{1+n^2}\times -\cos n\pi + \frac{1}{1+n^2}\right)\right]$$

$$\frac{1}{\pi}\left[\frac{1-e^{-\pi}}{1+n^2}\cos n\pi - \frac{e^{-\pi}+1}{1+n^2}\times\cos n\pi\right]$$

$$\frac{1}{\pi(1+n^2)}\left[1 - e^{-\pi}\cos n\pi - e^{-\pi}\cos n\pi + 1\right]$$

$$\frac{2}{\pi(1+n^2)}\left(1 - e^{-\pi}\cos n\pi\right)$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{0} e^x \sin nx\,dx + \int_{0}^{\pi} e^{-x}\sin nx\,dx$$

$$= \frac{1}{\pi}\left[\frac{e^x}{1+n^2}(\sin nx - n\cos nx)_{-\pi}^{0} + \frac{e^{-x}}{1+n^2}(-\sin nx - n\cos nx)_{0}^{\pi}\right]$$

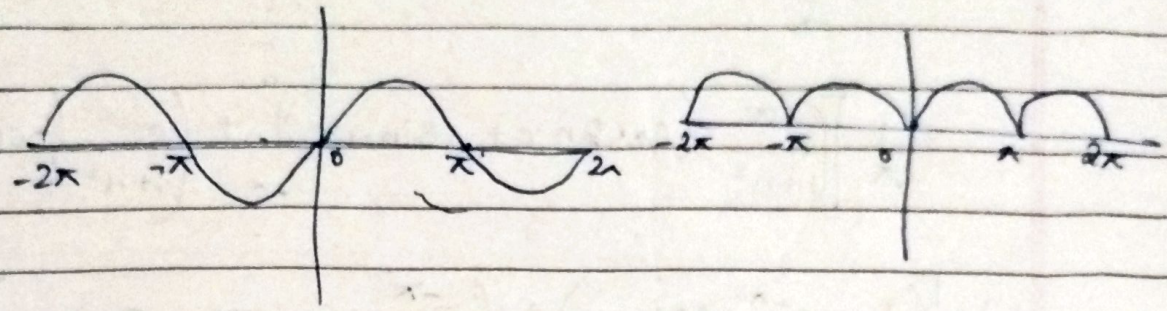
$$= \frac{1}{\pi}\left[\frac{1}{1+n^2}(-n\right.$$

$$\frac{1}{\pi}\left[\left(\frac{-n}{1+n^2} - \frac{e^{-\pi}}{1+n^2}\times\cos n\pi\right) + \left(e^{-\pi}\times - n\cos n\pi + n\right)\right]$$

$$\frac{1}{\pi}\left[\frac{-n + e^{-\pi}\cos n\pi}{1+n^2} - \frac{e^{-\pi}n\cos nx + n}{1+n^2}\right]$$

$$\frac{1}{\pi}[0] = 0$$

Q $f(x) = |\sin x| \quad -\pi < x < \pi$ $\cos(-\pi)$ $\cos\pi$



$$f(x) = \begin{cases} -\sin x & -\pi < x < 0 \\ \sin x & 0 < x < \pi \end{cases}$$

$$a_0 = \frac{1}{2\pi}\left[-\int_{-\pi}^{0} \sin x\,dx + \int_{0}^{\pi} \sin x\,dx\right]$$

$$= \frac{1}{2\pi}\left[-\left[(-\cos x)_{-\pi}^{0} + (-\cos x)_{0}^{\pi}\right]\right]$$

$$= \frac{1}{2\pi}\left[-\left(-(-1+1)\right) + \left(-(-1-1)\right)\right] = 0$$

$$= \frac{2+2}{\pi} = \frac{4}{\pi}$$

$$a_n = \frac{1}{\pi}\left[\left(\int_{-\pi}^{0} -\sin x \cos nx\,dx\right) + \left(\int_{0}^{\pi} \sin x \cos nx\,dx\right)\right]$$

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Q1 $f(x) = x^3 \quad 0 < x < 2\pi$

Q2 $f(x) = 1 + 2x \quad -\pi < x < \pi$

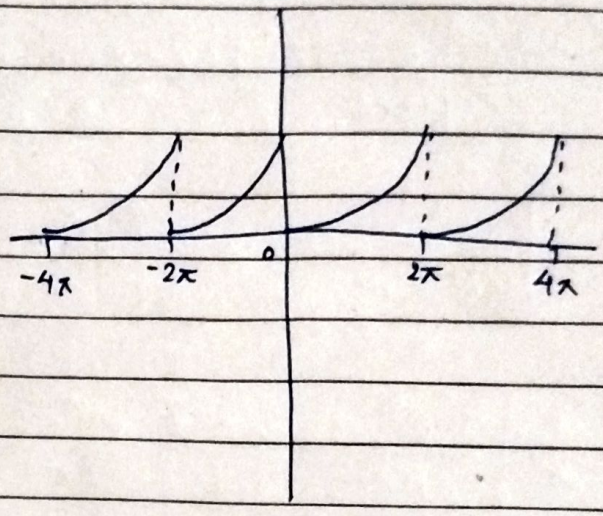
Q.1) $a_0 = \frac{1}{2\pi}\int_0^{2\pi} x^3\,dx$

$= \frac{1}{2\pi}\left[\frac{x^4}{4}\right]_0^{2\pi}$

$= \frac{1}{2\pi}\left[\frac{2\pi \times (2\pi)^3}{4}\right]$

$= \frac{8\pi^3}{4}$

$= \underline{\underline{2\pi^3}}$



$a_n = \frac{1}{\pi}\int_0^{2\pi} x^3 \cos(nx)\,dx$

$= \frac{1}{\pi}\left[\frac{x^3 \sin(nx)}{n} + \frac{3x^2 \cos(nx)}{n^2} - \frac{6x \sin(nx)}{n^3} - \frac{6\cos nx}{n^4}\right]_0^{2\pi}$

$a_n = \frac{1}{\pi}\left[\frac{12\pi^2}{n^2} - \frac{6}{n^4} + \frac{6}{n^4}\right]$

$= \frac{1}{\pi}\left[\frac{12\pi^2}{n^2}\right]$

$= \underline{\underline{\frac{12\pi}{n^2}}}$

$b_n = \frac{1}{\pi}\int_0^{2\pi} x^3 \sin(nx)\,dx$

$= \frac{1}{\pi}\left[\frac{-x^3\cos nx}{n} + \frac{3x^2 \sin nx}{n^2} + \frac{6x\cos nx}{n^3} - \frac{6\sin nx}{n^4}\right]_0^{2\pi}$

$$\frac{1}{\pi}\left[\frac{-(2\pi)^3}{n} + \frac{12\pi}{n^3}\right]$$

$$= \frac{1}{\pi}\left[\frac{12\pi}{n^3} - \frac{8\pi^3}{n}\right]$$

$$= \frac{12\pi}{n^3} - \frac{8\pi^2}{n}$$

$$x^3 = 2\pi^3 + \sum_{n=1}^{\infty}\left(\frac{12\pi}{n^2}\cos nx + \left(\frac{12\pi}{n^3} - \frac{8\pi^2}{n}\right)\sin nx\right).$$

Q.2) $f(x) = 1 + 2x \quad (-\pi < x < \pi)$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} 1 + 2x\,dx$$

$$= \frac{1}{2\pi}\left[x + x^2\right]_{-\pi}^{\pi}$$

$$= \frac{1}{2\pi}\left[\pi + \pi^2 + \pi - \pi^2\right]$$

$$= \frac{1}{2\pi}\left[2\pi\right] = 1$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi}(1+2x)\cos nx\,dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{\pi}\cos nx\,dx + 2\int_{-\pi}^{\pi} x\cos nx\,dx\right]$$

$$\int_{-\pi}^{\pi}\cos nx\,dx = \left[\frac{\sin nx}{n}\right]_{-\pi}^{\pi} = 0$$

$$2\int_{-\pi}^{\pi} x\cos nx\,dx = 2\left[\frac{x\sin nx}{n} + \frac{\cos nx}{n^2}\right]_{-\pi}^{\pi}$$

$$a_n = 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi}(1+2x)\sin nx\,dx$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi}\sin nx + 2\int_{-\pi}^{\pi}x\sin nx\,dx$$

$$= \frac{2}{\pi}\int_{-\pi}^{\pi}x\sin nx\,dx = \frac{2\times 2}{\pi}\int_{0}^{\pi}x\sin nx\,dx$$

$$= \frac{4}{\pi}\left[\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi}$$

$$b_n = \frac{4}{\pi}\left[-\frac{\pi\cos n\pi}{n}\right]$$

$$= -\frac{4\cos n\pi}{n}$$

Fourier Series With a constant period 2L

Let $f(x)$ be a periodic f^n with period 2L.

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right)$$

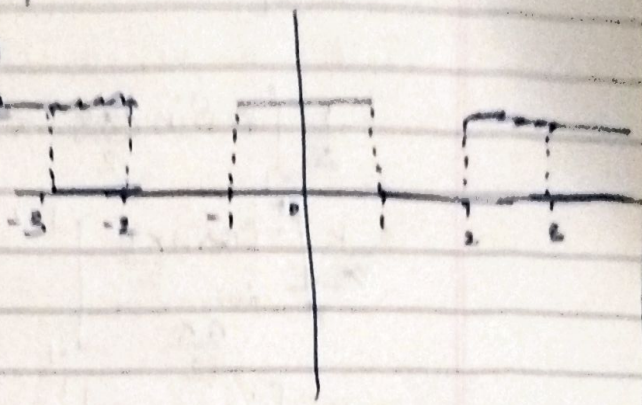
$$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx$$

Q. Draw the periodic graph:

$$f(x) = \begin{cases} 0, & -2 < x < -1 \\ K, & -1 < x < 1 \\ 0, & 1 < x < 2 \end{cases}$$



$$f(x+4) = f(x)$$

$$2L = 4$$

$$L = 2$$

$$a_0 = \frac{1}{4}\int_{-1}^{1} K\,dx$$

$$= \frac{1}{4}\left[Kx\right]_{-1}^{1}$$

$$= \frac{1}{4}2K + 2K = \frac{2AK}{A2} = \frac{2K}{2}$$

$$a_n = \frac{1}{2}\left[\int_{-1}^{1} K \frac{\cos n\pi x}{2}\,dx\right]$$

$$= \frac{K}{A^2}\int_{-1}^{1} \cos n\pi \frac{x}{L}\,dx$$

$$= \frac{K}{4}\left[\frac{\sin n\pi x}{n\pi}\right]_{-1}^{1} = \frac{K}{2}\left[\frac{\sin n\pi \frac{x}{L}}{\frac{n\pi}{L}}\right]_{-1}^{1}$$

$$= \frac{K}{4}\left[\frac{\sin n\pi - \cos n\pi}{n\pi \quad n\pi}\right] = \frac{K}{2}\left[\frac{\sin n\pi \frac{x}{2}}{\frac{n\pi}{2}}\right]_{-1}^{1}$$

$$= \frac{K \cdot \sin\frac{n\pi}{2} + \sin\frac{n\pi}{2}}{n\pi}$$

$$= \frac{2K \sin n\pi/2}{n\pi}$$

$$b_n = \frac{1}{2}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx$$

$$= \frac{1}{2}\int_{-1}^{1} K\sin\frac{n\pi x}{2}\,dx$$

$$= \frac{K}{2}\left[\frac{-\cos\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right]_{-1}^{1}$$

$$= \frac{K}{n\pi}\left[-\cos\frac{n\pi x}{2}\right]_{-1}^{1}$$

$$= \frac{K}{n\pi}\left[-\left(\cos\frac{n\pi}{2} - \cos\frac{n\pi}{2}\right)\right]$$

$$= \frac{-K}{n\pi}(0) = \underline{0}$$

$$\boxed{f(x) = \frac{K}{2} + \sum_{n=1}^{\infty}\left(\frac{2K}{n\pi}\sin\left(\frac{n\pi}{2}\right)\cdot\cos\frac{n\pi x}{2}\right)}$$

$$= \frac{K}{2} + \frac{2K}{\pi}\left(\cos\frac{\pi x}{2} - \frac{1}{3}\cos\frac{3\pi x}{2} + \frac{1}{5}\cos\frac{5\pi x}{2} \cdots\right)$$

Substitute $x = 0$ $f(x) = K$ therefore fourier series becomes:

$$K = \frac{K}{2} + \frac{2K}{\pi}\left(1 - \frac{1}{3} + \frac{1}{5} \cdots\right)$$

$$\frac{K}{2} = \frac{2K}{\pi}\left(1 - \frac{1}{3} + \frac{1}{5} - \cdots\right)$$

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$$

Q. $f(x) = x \quad , -L < x < L$

$$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx$$

$$= \frac{1}{2L}\left[\frac{x^2}{2}\right]_{-L}^{L}$$

$$= \frac{1}{2L}\left[\frac{L^2}{2} - \frac{L^2}{2}\right] = 0$$

$$a_n = \frac{1}{L}\int_{-L}^{L} x \cos\frac{n\pi x}{L}\,dx$$

$$= \frac{1}{L}\left[x\sin\frac{n\pi x}{L} - \right] = 0$$

$$b_n = \frac{1}{L}\int_{-L}^{L} x \sin\frac{n\pi x}{L}\,dx$$

$$= \frac{1}{L}\left[\frac{-x\cos\frac{n\pi x}{L}}{L/n\pi} + \frac{\sin\frac{n\pi x}{L}}{L/\left(\frac{n\pi}{L}\right)^2}\right]_{-L}^{L}$$

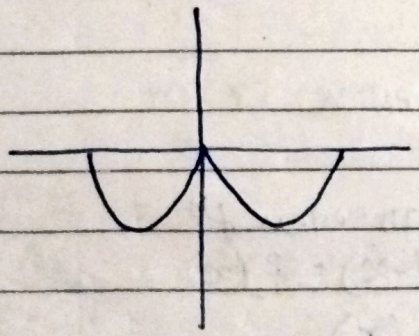
$$= \frac{1}{L}\left[\frac{-L\cos n\pi L}{\frac{n\pi}{L}} - \frac{\sin n\pi L}{L} + L\cos n\pi(-L) - \sin\frac{n\pi - L}{L}\right]$$

Even & Odd f^n

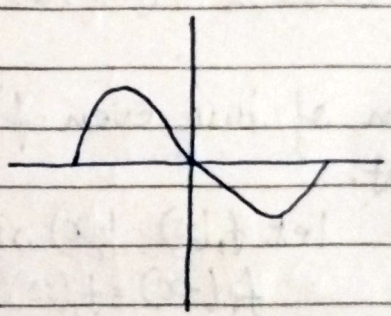
- A f^n is said to be even, if $f(-x) = f(x), \forall\ x \in D$
  A f^n is said to be odd, if $f(-x) = -f(x), \forall\ x \in D$

eg: $\cos(-x) = \cos x \Rightarrow$ even
$\sin(-x) = -\sin x \Rightarrow$ odd

Note: to check a f^n even or odd if it's defined to the left and rgt of the origin.
i.e it should be symmetric wrt x or y axis.

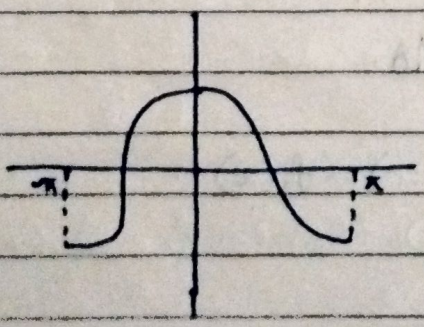


even f^n



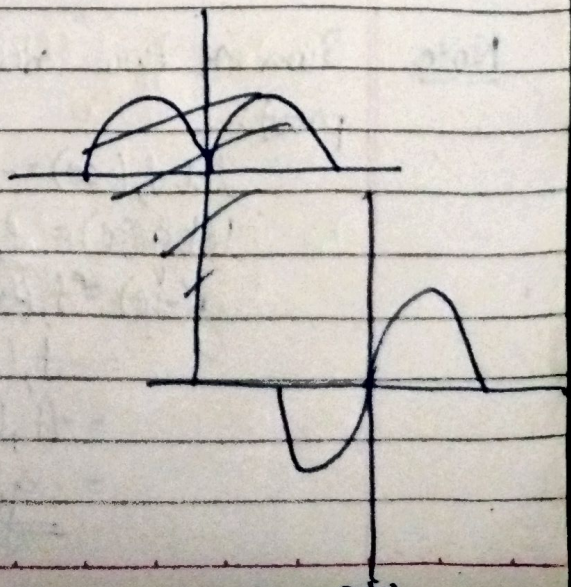
odd f^n

$f(x) = \cos x \quad -\pi < x < \pi$



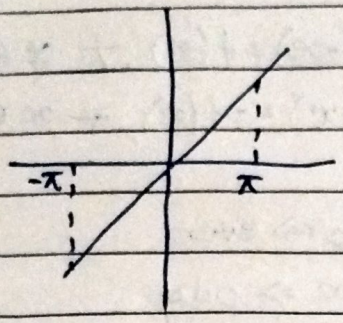
even

$f(x) = \sin x \quad -\pi < x < \pi$



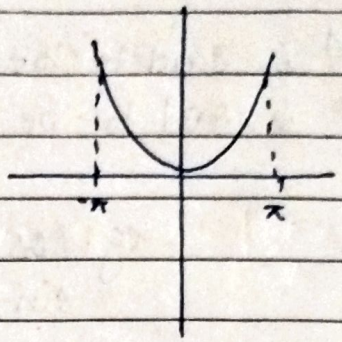
odd

$f(x) = x$



odd

$f(x) = x^2$



even

eg: $f(x) = x + x^2$

$f(-x) = -x + x^2$ i.e $x^2 - x$

that is $f(x)$ not eq to $-f(x)$ as $f(-x)$

Note Sum of two even f^n is even

proof:

let $f_1(x), f_2(x)$ are two even f^n

$f_1(-x) = f_1(x)$ & $f_2(-x) = f_2(x)$

let $g(x) = f_1(x) + f_2(x)$

$g(-x) = f_1(-x) + f_2(-x)$

$= f_1(x) + f_2(x)$

$= g(x) \Rightarrow$ even

Note Sum of two odd f^n is odd.

proof:

let $f_1(-x) = -f_1(x)$, $f_2(-x) = -f_2(x)$.

let $g(x) = f_1(x) + f_2(x)$

$g(-x) = f_1(-x) + f_2(-x)$

$= -f_1(x) - f_2(x)$

$= -(f_1(x) + f_2(x))$

$= -g(x) \Rightarrow$ odd

Note difference of two even f^n is even

let $g(x) = f_1 - f_2$

$g(-x) = f_1(-x) - f_2(-x)$

$= f_1(x) - f_2(x)$

$= g(x) \rightarrow$ even

Note ~~$g(x) = f_1 - f_2$~~

Note diff of two odd f^n is odd

let $g(x) = f_1 - f_2$

$g(-x) = f_1(-x) - f_2(-x)$

$= -f_1(x) + f(x)$

$-(f_1(x) - f_2(x)$

$= -g(x) \rightarrow$ odd

Note Product of two even f^n is even

$g(x) = f_1(x) \cdot f_2(x)$

$g(-x) = f_1(-x) \cdot f_2(-x)$

$= f_1(x) \cdot f_2(x) = g(x) =$ even

Note Product of two odd f^n is even

Note Product of odd f^n and even f^n is odd.

Fourier Series for Even & Odd f^n

Let $f(x)$ be a periodic f^n with period $2L$ then the fourier series of $f(x)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right]$$

$$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

RESULT: $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$, if $f(x)$ is even

0 if $f(x)$ is odd.

If $f(x)$ is even

$$a_0 = \frac{1}{2L}\int_{-a}^{a} f(x) = \frac{2}{2L}\int_{0}^{a} f(x)\,dx = \frac{1}{L}\int_{0}^{a} f(x)\,dx$$

$$a_n = \frac{1}{L}\int_{-a}^{a} f(x)\cos\left(\frac{n\pi x}{L}\right)dx = \frac{2}{L}\int_{0}^{a} f(x)\cos\left(\frac{n\pi x}{L}\right)dx$$

$b_n = 0$

Therefore if $f(x)$ is even series becomes

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

this is called fourier cosine series.

If $f(x)$ is odd

$$a_0 = 0$$

$$a_n = 0$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{2}{L}\int_{0}^{L} f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Therefor series becomes

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) dx$$

this is called fourier sine series.

Note: Evaluvating fourier cosine & sine series it is enough to define in the half range. It means that the f^n is periodic with period $2L$.

half range = $0 < x < L$

Note If $f(x)$ is 2π periodic

a) Fourier cosine series $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$

b) Fourier Sine Series $f(x) = \sum_{n=1}^{\infty} b_n \sin(nx)$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx)\, dx$$

Ex Find the fourier cosine & sine series of $f(x) = \cos x$, $0 < x < \pi$

Cosine series $= f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} \cos x\, dx$$

$$= \frac{[+\sin x]_0^{\pi}}{\pi} \, dx$$

$$= 0$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} \cos x \cos nx\, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \cos(1+n) + \cos(1-n)\, dx$$

$$= \frac{2}{\pi} \left[\frac{\sin(1+n)}{1+n} + \frac{\sin(1-n)}{1-n} \right]_0^{\pi} \quad n \neq 1$$

$$= 0$$

for $n = 1$

$$a_n = \frac{2}{\pi} \int_0^{\pi} \cos^2 x\, dx$$

$$= 1$$

Sine series

$$b_n = \frac{2}{\pi}\int_0^{\pi} \cos x \sin nx \, dx$$

$$= \frac{1}{\pi}\int_0^{\pi} \sin(n+1)x + \sin(n-1)x \, dx$$

$$= \frac{-1}{\pi}\left(\frac{\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1}\right)_0^{\pi}$$

$$= \frac{-1}{\pi}\left[\frac{\cos(n+1)\pi}{n+1} + \frac{\cos(n-1)\pi}{n-1} - \frac{1}{n+1} - \frac{1}{n-1}\right]$$

$$= \frac{-1}{\pi}\left[\frac{(-1)^{n+1}}{n+1} + \frac{(-1)^{n-1}}{n-1} - \frac{2n}{n^2-1}\right]$$

$n=2$, $b_1 = \frac{-1}{\pi}\left[\frac{-1}{3} + 1 - \frac{4}{3}\right]$

$$= \frac{-1}{\pi}\left[\frac{-5+1}{3}\right]$$

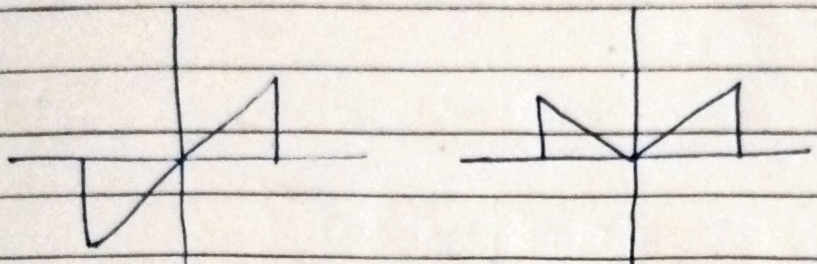
$$\frac{-1}{\pi}\,\frac{-5+3}{3}$$

$$= \frac{2}{3\pi}$$

$n=3=0$

$n=4$.

Half Range Expansion
(even and odd extensions)



odd extension
→ Sine series

even extension.
→ cosine series

Let $f(x)$ be defined in the half range $0<x<L$ then $f(x)$ can be extended to $-L<x<0$ either by even extension or by odd extension. So that $f(x)$ is periodic with period $2L$. For even extension we get a fourier cosine series $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{L}\, dx$

here $a_0 = \frac{1}{L}\int_0^L f(x)\,dx$

$a_n = \frac{2}{L}\int_0^L f(x) \cos\frac{n\pi x}{L}\, dx$

For odd extension we get a fourier sine series

$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{L}$

$b_n = \frac{2}{L}\int_0^L f(x) \sin\frac{n\pi x}{L}\, dx.$

eg: find the half range expansions of $f(x) = x \quad 0 < x < 1$

$$a_0 = \frac{1}{L}\int_0^L f(x)\,dx$$

$$= \int_0^1 x\,dx$$

$$\left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}$$

$$a_n = \frac{2}{2}\int_0^1 x\cos n\pi x\,dx$$

$$2\left[x\frac{\sin n\pi x}{n\pi} - 1x - \frac{\cos n\pi x}{(n\pi)^2}\right]_0^1$$

$$2\left[\frac{\cos n\pi}{(n\pi)^2} - \frac{1}{(n\pi)^2}\right]$$

$$\frac{2}{n^2\pi^2}\left((-1)^n - 1\right)$$

$$a_n = \begin{cases} 0 & n \text{ even} \\ \frac{-4}{n^2\pi^2} & \text{odd} \end{cases}$$

$$b_n = \frac{2}{L}\int_0^L f(x)\sin n\pi x\,dx$$

$$= 2\int_0^1 x\sin n\pi x\,dx$$

$$\left[x\frac{-\cos n\pi x}{n\pi} - 1x\frac{\sin n\pi x}{n\pi^2}\right]_0^1$$

$$\left[\frac{-x\cos n\pi x}{n\pi} + \frac{\sin n\pi x}{n\pi^2}\right]_0^1$$

$$= 2\left[\frac{-(-1)^n}{n\pi}\right]$$

Even extension

$$f(x) = \frac{1}{2} + \sum_{n=1,3,5\ldots}^{\infty} \frac{-4}{n^2\pi^2} \cos n\pi x$$

Odd extension.

$$f(x) = \sum_{n=1}^{\infty} -2\left[\frac{(-1)^n}{n\pi}\right] \sin n\pi x$$

Find the half range expansion of

$$f(x) = \begin{cases} \frac{2Kx}{L} & 0 < x < \frac{L}{2} \\ \frac{2K}{L}(L-x) & \frac{L}{2} < x < L \end{cases}$$

$$a_0 = \frac{1}{L}\left[\int_0^{L/2} \frac{2Kx}{L}\,dx + \int_{L/2}^{L} \frac{2K}{L}(L-x)\,dx\right]$$

$$= \frac{1}{L}\left[\left[\frac{Kx^2}{L}\right]_0^{L/2} + \left[\frac{2KLx}{L} - \frac{2Kx^2}{L}\right]_{L/2}^{L}\right]$$

$$= \frac{1}{L}\left[\frac{KL^2}{4L} + \left[\frac{2KL^2}{L} - \frac{KL^2}{L} - \frac{2KL^2}{2L} + \frac{KL^2}{4L}\right]\right]$$

$$= \frac{1}{L}\left[\frac{KL^2}{4L} + \frac{2KL^2}{L} - \frac{KL^2}{L} - \frac{2KL^2}{L} + \frac{KL^2}{4L}\right]$$

$$= \frac{1}{L}\left[\frac{2KL^2}{2L}\right] = \frac{K}{2}$$

$$a_n = \frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}\,dx$$

$$= \frac{2}{L}\left[\int_0^{L/2}\frac{2Kx}{L}\cos\frac{n\pi x}{L} + \int_{L/2}^{L}\frac{2K}{L}(L-x)\cos\frac{n\pi x}{L}\right]$$

$$\frac{2\times 2K}{L\ L}\int_0^{L/2} x\cos\frac{n\pi x}{L} + \int_{L/2}^{L}(L-x)\cos n\pi x$$

$$\frac{4K}{L^2}\left[x\cdot\sin\frac{n\pi x}{L}\times\frac{L}{n\pi} - 1\times-\cos\frac{n\pi x}{L}\times\frac{L^2}{(n\pi)^2}\right]_0^{L/2}$$

$$+\left[(L-x)\times\frac{\sin n\pi x}{L}\times\frac{L}{n\pi} - (-1)\times\frac{-\cos n\pi x}{L}\times\frac{L^2}{(n\pi)^2}\right]_{L/2}^{L}$$

$$= \frac{4K}{L^2}\left[x\sin\frac{n\pi x}{L}\times\frac{L}{n\pi} + \cos\frac{n\pi x}{L}\times\frac{L^2}{(n\pi)^2}\right]_0^{L/2} +$$

$$\frac{4K}{L^2}\left[(L-x)\sin\frac{n\pi x}{L}\times\frac{L}{n\pi} - \cos\frac{n\pi x}{L}\times\frac{L^2}{(n\pi)^2}\right]$$

$$\frac{4K}{L^2}\left[\frac{L}{2}\sin\frac{n\pi}{2}\times\frac{L}{n\pi} + \left(\frac{L^2}{(n\pi)^2}\right)\cos\frac{n\pi}{2} - \left(\frac{L}{n\pi}\right)^2 - \cos n\pi\left(\frac{L}{n\pi}\right)^2 - \right.$$

$$\left.\frac{L}{2}\sin\left(\frac{n\pi}{2}\right)\cdot\frac{L}{n\pi} + \cos\frac{n\pi}{2}\left(\frac{L}{n\pi}\right)^2\right]$$

$$= \frac{4K}{L^2}\left(2\left(\frac{L}{n\pi}\right)^2\cos\frac{n\pi}{2} - \left(\frac{L}{n\pi}\right)^2\cos n\pi - \left(\frac{L}{n\pi}\right)^2\right]$$

$$= \frac{4K}{L^2}\left(\left(\frac{L}{n\pi}\right)^2\left(2\cos\left(\frac{n\pi}{2}\right) - \cos(n\pi) - 1\right)\right.$$

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Q. Verify $f(x) = x^3 - 2x$ is even or odd.
$-1 < x < 1$ ~~$-\pi < x < \pi$~~, $f(x+2) = f(x)$
hence find its fourier series.

Q. Obtain the half range expansion of
$$f(x) = \begin{cases} 0, & 0 < x < L/2 \\ 1-x, & L/2 < x < L \end{cases}$$

Q. Find the fourier series expansion of $f(x) = x(\pi - |x|)$, $-\pi < x < \pi$

1) $f(-x) = -x^3 + 2x = -(f(x))$
hence odd fn.

$a_0 + a_n = 0$

$$b_n = \frac{2}{2L}\int_0^L f(x) \sin\frac{n\pi x}{L}\, dx$$

$$= 2\int_0^1 (x^3 - 2x) \sin n\pi x\, dx$$

$$= 2\left[\int_0^1 x^3 \sin n\pi x - \int_0^1 2x \sin n\pi x\, dx\right]$$

$$2\left[(x^3-2x)\frac{-\cos n\pi x}{n\pi} - (3x^2-2)\frac{-\sin n\pi x}{(n\pi)^2} + (6x)\frac{\cos n\pi x}{(n\pi)^3} - 6\frac{\sin n\pi x}{(n\pi)^4}\right]_0^1$$

$$= 2\left(\frac{\cos n\pi}{n\pi} + 6\frac{\cos n\pi}{(n\pi)^3}\right)$$

$$= \frac{2\cos n\pi}{(n\pi)^3}\left((n\pi)^2 + 6\right)$$

fourier sin series.

$$\sum_{n=1}^{\infty} \frac{2\cos(n\pi)\left((n\pi)^2+6\right)}{(n\pi)^3} \cdot \sin n\pi x$$

a) $a_0 = \frac{1}{L}\left[\int_0^{L/2} 0\, dx + \int_{L/2}^{L} (L-x)\, dx\right]$

$= \frac{1}{L}\left(Lx - \frac{x^2}{2}\right)_{L/2}^{L}$

$= \frac{1}{L}\left(L^2 - \frac{L^2}{2} - \frac{L^2}{2} + \frac{L^2}{4}\right)$

$= \frac{1}{L}\left[\frac{4L^2}{4}\right] = \frac{1}{L}\left(\frac{L^2}{2} - \frac{L^2}{4}\right)$

$= L^2/4$ // $= \frac{L}{2} - \frac{L}{4} = \frac{4L-2L}{8} = \frac{2L}{8} = \frac{L}{4}$

$a_n = \frac{2}{L}\left[(L-x) \times \sin\frac{n\pi x}{L} \times \frac{L}{n\pi} - x - 1x - \cos\frac{n\pi x}{L}\left(\frac{L}{4\pi}\right)^2\right]_{L/2}^{L}$

$= \frac{2}{L}\left[\cos(n\pi)\left(\frac{L}{4\pi}\right)^2 + \cos\left(\frac{n\pi}{2}\right)\left(\frac{L}{n\pi}\right)^2 - \frac{L}{2} \times \frac{L}{n\pi} \cdot \sin\frac{n\pi}{2}\right]_{L/2}$

$= \frac{2}{L} \times \left(\frac{L}{n\pi}\right)^2 \left[\frac{L\cos(n\pi)}{n\pi} + L\cos\left(\frac{n\pi}{2}\right)\right] - \frac{L}{2}\sin\frac{n\pi}{2}$

$= \frac{2L}{(n\pi)^2}\left[\cos\left(\frac{n\pi}{2}\right) - \cos(n\pi)\right]$

$\frac{2}{n\pi}\left[\frac{-L}{n\pi}\cos n\pi - \frac{L}{2}\sin\frac{n\pi}{2} + \frac{L}{n\pi}\cos\left(\frac{n\pi}{2}\right)\right]$

$\frac{2L}{n\pi}\left[\frac{-\cos n\pi}{n\pi} - \frac{1}{2}\sin\frac{n\pi}{2} + \frac{1}{n\pi}\cos\frac{n\pi}{2}\right]$

$$b_n = \frac{2}{L}\int_0^L f(x)\,\sin\frac{n\pi x}{L}\,dx$$

$$= \frac{2}{L}\int_{L/2}^{L}(L-x)\,\sin\frac{n\pi x}{L}\,dx$$

$$= \frac{2}{L}\left[(L-x)\left(-\cos\frac{n\pi x}{L}\right)\times\frac{L}{n\pi} - (-1)\left(-\sin\frac{n\pi x}{L}\right)\times\left(\frac{L}{n\pi}\right)^2\right]_{L/2}^{L}$$

$$\frac{2}{L}\left[\frac{L}{2}\times - \cos\frac{n\pi}{2}\times\frac{L}{n\pi} - \sin\left(\frac{n\pi}{2}\right)\times\left(\frac{L}{n\pi}\right)^2\right]$$

$$\frac{2}{n\pi}\left[\frac{-L\cos\left(\frac{n\pi}{2}\right)}{} - \frac{L\sin\left(\frac{n\pi}{2}\right)}{n\pi}\right]$$

$$\frac{-2L}{n\pi}\left[\cos\left(\frac{n\pi}{2}\right) + \frac{\sin\left(\frac{n\pi}{2}\right)}{n\pi}\right]$$

Parseval Identity

One of the most important result in fourier series The total energy of a function is equal to the total energy of its fourier coeff.

$$\frac{1}{\pi}\int_{-\pi}^{\pi}[f(x)]^2\,dx = 2a_0^2 + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)$$

1. Consider the periodic f^n

$$f(x) = \begin{cases} \pi + x, & -\pi < x < 0 \\ \pi - x, & 0 < x < \pi \end{cases}$$

The fourier series expansion of $f(x)$ is given by.

$$f(x) = \frac{\pi}{2} + \frac{4}{\pi}\left\{\cos x + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots\right\}$$

$$a_0 = \frac{\pi}{2},\quad b_n = 0,\quad a_n = \begin{cases} \frac{4}{n^2\pi}, & n \text{ is odd} \\ 0, & n \text{ is even} \end{cases}$$

By parsevals identity;

$$2a_0^2 + \sum_{n=1}^{\infty}(a_n^2 + b_n^2) = \frac{1}{\pi}\int_{-\pi}^{\pi}[f(x)]^2\,dx$$

substituting a_0, a_n, b_n, we have.

$$2\left(\frac{\pi}{2}\right)^2 + \frac{16}{\pi^2} + \frac{16}{3^4\pi^2} + \frac{16}{5^4\pi^2} + \cdots = \frac{1}{\pi}\int_{-\pi}^{\pi} f[(x)]^2\,dx$$

$$\frac{\pi^2}{2} + \frac{16}{\pi^2}\left(1 + \frac{1}{3^4} + \frac{1}{5^4} + \cdots\right) = \frac{1}{4}\int_{-\pi}^{\pi}[f(x)]^2\,dx$$

$$\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)^2\,dx = \frac{1}{\pi}\left\{\int_{-\pi}^{0}(\pi+x)^2\,dx + \int_{0}^{\pi}(\pi-x)^2\,dx\right\}$$

$$= \frac{1}{\pi}$$

Q. $f(x) = \sin x$ from $-\pi$ to π odd f^n

$a_0 = 0,\ a_n = 0,\ b_1 = 1,\ b_n = 0\ (n \neq 1)$

passeval:

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \sin^2 x\,dx = \frac{1}{\pi}\,\pi = 1$$

$$0 + 1^2 = 1$$

$$1 = 1$$

Q. $f(x) = \cos x$ from $-\pi$ to π even f^n

$a_1 = 1,\ a_0 = 0,\ b_n = 0$

passeval: $\frac{1}{\pi}\int_{-\pi}^{\pi} \cos^2 x\,dx = 1$

example 2: $f(x) = \begin{cases} 0, & -\pi < x < -\pi/2 \\ 1, & -\pi/2 < x < \pi/2 \\ 0, & \pi/2 < x < \pi \end{cases}$; $f(x+2\pi) = f(x)$

Show that the fourier series expansion of $f(x) = \frac{1}{2} + \frac{2}{\pi}\left\{\cos x - \frac{1}{3}\cos 3x + \frac{1}{5}\cos 5x - \cdots\right\}$

hence use parseval to verify that

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$$

Fourier Integral

$$\text{Fourier } f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$\text{as } L \to \infty, \quad f(x) = \int_0^{\infty} A_w \cos wx + B_w \sin wx \, dw$$

$$A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos(wx) \, dx$$

$$B(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin(wx) \, dx$$

One way of extending the method of fourier analysis to non periodic that are defined on $(-\infty, \infty)$, consider a portion of the function $f(x)$ over some finite interval and then extended this f^n over the integral $(-\infty$ to $\infty)$.

In $f(x)$ $\frac{n\pi x}{L}$ is discrete frequencies, now increase the period $L = 10, 100, 1000$ and when $L \to \infty$ the f^n is no longer periodic and $\frac{\pi}{L}$ becomes smaller i.e closer to 0.

i.e discrete freq mearge to a continous spectrum summation changes to integral called fourier integral.

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Fourier Cosine integral

If $f(x)$ is an even f^n.

$$A(w) = \frac{2}{\pi}\int_0^\infty f(x)\cos(wx)\,dx \quad , B(w) = 0$$

Fourier Cosine integral $= f(x) = \int_0^\infty A(w)\cos(wx)\,dx$

If $f(x)$ is an odd f^n

$$A(w) = 0 \quad , B(w) = \frac{2}{\pi}\int_0^\infty f(x)\sin(wx)\,dx$$

Fourier Sine integral $= f(x) = \int_0^\infty B(w)\sin(wx)\,dx$

Note 1: Thus Fourier integral is a non periodical signal made up infinitely many sine and cosine waves.

Note 2: In fourier series a_0, a_n, b_n are constants in fourier integral $A(w), B(w)$ are functions of w.

Comparison

F.S.	F.I.
Periodic	Non periodic
Finite integral	Infinite integral
Discrete freq	Continous freq
Summation	Integration
Const. Coeff	Spectral coeff
a_0, a_n, b_n	$A(w), B(w)$

1) $f(x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$

and evaluate $\int_0^\infty \frac{\sin w}{w} dw$.

$$A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos wx \, dx$$

$$= \frac{1}{\pi} \int_{-1}^{1} 1 \cdot \cos wx \, dx$$

$$= \frac{1}{\pi} \left(\frac{\sin wx}{w} \right)_{-1}^{1}$$

$$= \frac{2}{\pi} \frac{\sin w}{w}$$

$$B(w) = \frac{1}{\pi} \int_{-1}^{1} 1 \cdot \sin wx \, dx = 0$$

Fourier integral

$$f(x) = \int_0^\infty A(w) \cos wx + B(w) \sin wx \, dw$$

$$f(x) = \int_0^\infty \frac{2}{\pi} \frac{\sin w}{w} \cdot \cos wx \, dw$$

$$\int_0^\infty \frac{\sin w}{w} \cos wx \, dw = \frac{\pi}{2} f(x)$$

$$= \begin{cases} \pi/2 \times 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$$

at $x = 0$, $f(0) = 1$

$$\int_0^\infty \frac{\sin w}{w} \cos 0 \, dw = \frac{\pi}{2} \cdot f(0)$$

$$\int_0^\infty \frac{\sin w}{w} dw = \frac{\pi}{2} //$$

2) $f(x) = e^{-kx}$, $k>0$, $x>0$

F.C.I

$$A\omega = \frac{2}{\pi}\int_0^\infty f(x)\cos\omega x\,dx$$

$$= \frac{2}{\pi}\int_0^\infty e^{-kx}\cos\omega x\,dx$$

$$\frac{2}{\pi}\left[\frac{e^{-kx}}{k^2+\omega^2}\left(-k\cos\omega x + \omega\sin\omega x\right)\right]_0^\infty$$

$$= \frac{2k}{\pi(k^2+\omega^2)}$$

$$FCI = \int_0^\infty \frac{2k}{\pi(k^2+\omega^2)}\cdot\cos\omega x\,d\omega = fx$$

$$\int_0^\infty 2\,\frac{\cos\omega x}{(k^2+\omega^2)} = \frac{\pi}{2k} f(x)$$

$$= \frac{\pi}{2k} e^{-kx} \quad x>0, k>0$$

FSI =

$$B(\omega) = \frac{2}{\pi}\int_0^\infty e^{-kx}\sin\omega x\,dx$$

$$= \frac{2}{\pi}\left[\frac{e^{-kx}}{k^2+\omega^2}\left(-k\sin\omega x + \omega\cos\omega x\right)\right]_0^\infty$$

$$= \frac{2}{\pi}\left[\frac{\omega}{k^2+\omega^2}\right]$$

$$FSI = \int_0^\infty B(\omega)\sin\omega x\,d\omega$$

$$= \frac{2}{\pi}\int_0^\infty \frac{\omega}{k^2+\omega^2}\sin\omega x\,dx = \frac{\pi f(x)}{2} \;;\; \frac{\pi}{2}e^{-kx},\ k>0,\ x>0.$$

Show that

① $$\int_0^\infty \frac{\cos wx + w \sin wx}{1+w^2}\, dw = \begin{cases} 0, & x<0 \\ \pi/2, & x=0 \\ \pi e^{-x}, & x>0 \end{cases}$$

Prove that

$$A(w) = \frac{1}{1+w^2}$$

$$B(w) = \frac{w}{1+w^2}$$

Proof

$$f(x) = \pi e^{-x}$$

$$A(w) = \frac{1}{\pi}\int_0^\infty \pi e^{-x} \cdot \cos wx\, dx$$

$$= \left[\frac{e^{-x}}{1+w^2}\left(-\cos(wx) + w \sin wx\right)\right]_0^\infty$$

$$= \frac{1}{1+w^2}$$

$$B(w) = \frac{1}{\pi}\int_0^\infty \pi e^{-x} \sin wx\, dx$$

$$= \left[\frac{e^{-x}}{1+w^2}\left(-\sin wx - w\cos wx\right)\right]_0^\infty$$

$$= \frac{w}{1+w^2}$$

~~Hence $\int_0^\infty \cos wx$~~

Hence $$\int_0^\infty \frac{1}{1+w^2}\cos wx + \frac{w}{1+w^2}\sin wx\, dw = f(x)$$

$$= \begin{cases} 0, & x<0 \\ \pi/2, & x=0 \\ \pi e^{-x}, & x>0. \end{cases}$$

2. $$\int_0^\infty \frac{\sin w \cos wx}{w}\, dw = \begin{cases} \pi/2, & 0<x<1 \\ \pi/4, & x=1 \\ 0, & x>1 \end{cases}$$

It is a fourier cosine integral:

$$A(w) = \frac{2}{\pi}\int_0^\infty f(x)\cos wx\, dx$$

$$= \frac{2}{\pi}\int_0^1 \frac{\pi}{2}\cos wx\, dx$$

$$\left(\frac{\sin wx}{w}\right)_0^1$$

$$A(w) = \frac{\sin w}{w}$$

$$FCI = \int_0^\infty \frac{\sin w}{w}\cdot \cos(wx)\, dw = f(x) = \begin{cases} \frac{\pi}{2}, & 0<x<1 \\ \frac{\frac{\pi}{2}+0}{2} = \frac{\pi}{4}, & x=1 \\ 0, & x>1 \end{cases}$$

3.) $$\int_0^\infty \frac{1-\cos(\pi w)}{w}\sin wx\, dw = \begin{cases} \pi/2, & 0<x<\pi \\ 0, & x>\pi \end{cases}$$

It is a fourier sine integral.

$$B(w) = \frac{2}{\pi}\int_0^\infty \frac{\pi}{2}\sin wx\, dx$$

$$\int_0^\pi \sin wx\, dx$$

$$B(w) = \left[\frac{-\cos wx}{w}\right]_0^\pi = \frac{1-\cos \pi x}{w}$$

FSI $= \int_0^\infty \frac{1-\cos\omega x}{\omega} \sin\omega x \, d\omega = \begin{cases} \pi/2, & 0<x<\pi \\ \pi/4 & x=\pi \\ 0, & x>\pi \end{cases}$

Q.4 Find the fourier cosine integral of

① $f(x) = \frac{1}{1+x^2}$ $x>0$

② $f(x) = \begin{cases} \sin x & 0<x<\pi \\ 0 & \text{otherwise} \end{cases}$

Q.5 Find the fourier sine integral of

① $f(x) = \begin{cases} a^2-x^2, & 0<x<a \\ 0, & \text{otherwise} \end{cases}$

② $f(x) = \begin{cases} x, & 0<x<a \\ 0, & \text{otherwise} \end{cases}$

Q.4)) $f(x) = \frac{1}{1+x^2}$ $x>0$

$A(\omega) = \frac{2}{\pi}\int_0^\infty \frac{1}{1+x^2}\cos\omega x \, dx$

$= \frac{2}{\pi}$

Fourier Transform

Fourier transform expresses an non periodic f^n in terms of a continus spectrum of freq. $f(x)$ is defined on $(-\infty, \infty)$ and is absolutely integrable.
i.e $\int_{-\infty}^{\infty} |f(x)| < \infty$ is finite.

Fouries transform is obtained from fourier integral.
Fouries sin transform is obtained from fouries sin integral where $f(x)$ is defined on $(0, \infty)$.
Fouries cosine transform is obtained from fourier cosine integral where $f(x)$ is defined on $(0, \infty)$.
The values of the transform tells the strenght of each waves, low freq. it is smooth and slowly varing. High freq means theis is a rapid chage.

Thus fouries transforms decomped the orignal function into simple freq curves and analyske the strenght of the freq.

Fouries sine transform: (F.S.T)

$$FSI = \int_0^{\infty} B(w) \sin wx \, dw. \quad \text{——(1)}$$

$$B(w) = \frac{2}{\pi} \int_0^{\infty} f(x) \sin wx \, dx \quad \text{——(2)}$$

$$\text{Setting } B(w) = \sqrt{\frac{2}{\pi}} \hat{f}_s(w) \quad \text{——(3)}$$

Substitute (3) in (2)

$$\sqrt{\frac{2}{\pi}}\,\hat{f}_s(w) = \frac{2}{\pi}\int_0^\infty f(x)\sin wx\,dx$$

$$\boxed{\hat{f}_s(w) = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\sin(wx)\,dx.}$$ Fourier Sine Transform

Substitute (3) in (1)

$$\boxed{f(x) = \sqrt{\frac{2}{\pi}}\int_0^\infty \hat{f}_s(w).\sin(wx)\,dw.}$$ Inverse fourier Sine transform.

Fourier Cosine Transform (FCT)

FCI, $f(x) = \int_0^\infty A(w)\cos wx\,dw$ — (1)

$A(w) = \frac{2}{\pi}\int_0^\infty f(x)\cos wx\,dx$ — (2)

Let $A(w) = \sqrt{\frac{2}{\pi}}\,\hat{f}_c(w)$ —— (3)

Substitute (3) in (2).

$$\sqrt{\frac{2}{\pi}}\,\hat{f}_c(w) = \frac{2}{\pi}\int_0^\infty f(x)\cos wx\,dx$$

$$\boxed{\hat{f}_c(w) = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\cos wx\,dx}$$ Fourier Cosine transform.

Substitute ③ in ①

$$f(x) = \int_0^\infty \sqrt{\frac{2}{\pi}}\, \hat{f}_c(w) \cos(wx)\, dw$$

$$\boxed{f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_c(w) \cos(wx)\, dw}$$ Inverse fourier cosine transform.

NOTE: FCT and FST are linear.
for this we will have to prove

$$\boxed{\hat{f}_c(af+bg) = a\hat{f}_c + b\hat{g}_c}$$

$$\hat{f}_c(af+bg) = \sqrt{\frac{2}{\pi}} \int_0^\infty (af+bg) \cos wx\, dx$$

$$= \sqrt{\frac{2}{\pi}} \left(a\int_0^\infty f \cos wx\, dx + b\int_0^\infty g \cos wx\, dx \right)$$

$$= a\hat{f}_c(w) + b\hat{g}_c(w)$$

Similarly, $\hat{f}_c(af+bg) = a\hat{f}_s(w) + b\hat{g}_s(w)$

Q. Find FST & FCT of

$$f(x) = \begin{cases} K, & 0 < x < a \\ 0, & x > a \end{cases}$$

$$F.S.T : \hat{f}_s(w) = \sqrt{\frac{2}{\pi}} \int_0^a K \sin(wx)\, dx$$

$$= -\sqrt{\frac{2}{\pi}} K \left(\frac{\cos wx}{w} \right)_0^a$$

$$= -\sqrt{\frac{2}{\pi}} \frac{K}{w} (\cos wa - 1)$$

$$F.C.T : \hat{f}_c(w) = \sqrt{\frac{2}{\pi}} \int_0^a K \cos(wx)\, dx$$

$$= \sqrt{\frac{2}{\pi}} \frac{K}{w} \left(\sin wx \right)_0^a$$

$$= \sqrt{\frac{2}{\pi}} \frac{K}{w} \sin wa$$

NOTE: The value represents how much are present in the signal ↳ how much freq. and it measures the strength of freq.

H.W find FCT & FST of $f(x) = \begin{cases} x^2, & 0 < x < 1 \\ 0, & x > 1 \end{cases}$

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Cosine & Sine transforms derivatives.

Let $f(x)$ be continous and absolutely integrable on the x axis and $f'(x)$ be piecewise continous on each interval, $f(x) \to 0$ & $x \to \infty$,

The FCT of the derivative ie $f_c(f'(x))$

$$f_c(f'(x)) = \omega f_s(f(x)) - \sqrt{\frac{2}{\pi}} f^s(0)$$

$$f_s(f'(x)) = -\omega f_c(f(x))$$

FCT. $\hat{f}_c(f) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos \omega x \, dx$

$$\hat{f}_c(f') = \sqrt{\frac{2}{\pi}} \int_0^\infty \underset{\text{II}^{nd}}{f'(x)} \underset{\text{I}^{st}}{\cos \omega x} \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left[[\cos \omega x \cdot f(x)]_0^\infty - \int_0^\infty -\omega \sin \omega x \cdot f(x) \right] dx$$

$$= \sqrt{\frac{2}{\pi}} \left(0 - f(0) \right) + \sqrt{\frac{2}{\pi}} \cdot \omega \int_0^\infty f(x) \sin \omega x \, dx$$

as $x \to \infty$, $f(x) \to 0$

$$= -\sqrt{\frac{2}{\pi}} f(0) + \omega \hat{f}_s f(x)$$

FST: $\hat{f}_s(f'(x)) = \sqrt{\frac{2}{\pi}} \int_0^\infty f'(x) \sin \omega x \, dx$

$= \sqrt{\frac{2}{\pi}} \left[(\sin \omega x \cdot f(x))_0^\infty - \int_0^\infty \omega \cos \omega x \cdot f(x) \, dx \right]$

$= \sqrt{\frac{2}{\pi}} (0-0) - \omega \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos \omega x \, dx$

$= -\omega \hat{f}_c(f(x))$

2nd derivative

FCT: $\hat{f}_c(f''(x)) = \omega \hat{f}_s(f'(x)) - \sqrt{\frac{2}{\pi}} f'(0)$

$= \omega x - \omega \hat{f}_c(f(x)) - \sqrt{\frac{2}{\pi}} f'(0)$

$\hat{f}_c(f''(x)) = -\omega^2 \hat{f}_c(f(x)) - \sqrt{\frac{2}{\pi}} f'(0)$

FST: $\hat{f}_s(f''(x)) = -\omega \hat{f}_c(f'(x))$

$= -\omega \left[\omega \hat{f}_s(f(x)) - \sqrt{\frac{2}{\pi}} f(0) \right]$

$= -\omega^2 \hat{f}_s(f(x)) + \omega \sqrt{\frac{2}{\pi}} f(0)$

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Q1) Find inverse FCT of $f(w) = e^{-w}$.

Q2) Find inverse FST of $f(w) = \frac{1}{w} e^{-aw}$

Ans 1. $f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_c(w) \cos wx \, dw$

$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-w} \cos wx \, dw$

$a = -1, b = x$

$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-w}}{1+x^2} \left(-\cos wx + x \sin wx \right) \right]_0^\infty$

$= \sqrt{\frac{2}{\pi}} \times \frac{1}{1+x^2} \left(e^{-w} \left(-\cos wx + x \sin wx \right) \right)_0^\infty$

$= \sqrt{\frac{2}{\pi}} \times \frac{1}{1+x^2} (0+1)$

$= \underline{\underline{\sqrt{\frac{2}{\pi}} \frac{1}{1+x^2}}}$

Ans 2 $f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_s(w) \sin wx \, dw$

$= \sqrt{\frac{2}{\pi}} \left[\int_0^\infty \frac{1}{w} e^{-aw} \sin wx \, dw \right]$

differentiate both side wrt x.

$f'(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{1}{w} e^{-w} \cos wx \, dw$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-a\omega} \cos\omega x \, d\omega$$

$$f(x) = \sqrt{\frac{2}{\pi}} \left[\frac{e^{-a\omega}}{a^2+x^2} (-a\cos\omega x + x \sin\omega x) \right]_0^\infty$$

$$= \sqrt{\frac{2}{\pi}} \frac{a}{a^2+x^2}$$

$$\therefore f(x) = \sqrt{\frac{2}{\pi}} \int \frac{a}{a^2+x^2} dx$$

$$= \sqrt{\frac{2}{\pi}} \tan^{-1}\left(\frac{x}{a}\right) + C$$

Q.1 Find the cosine and sine series $f(x) = 1+x$, $0 < x < \frac{1}{2}$

Q.2 Determine the inverse fourier cosine transform of $\frac{2}{4+\omega^2}$.

Q.3 Determine the fourier integral of $f(x) = e^{-3x}$, $x > 0$.